

WIRELESS NETWORKING

Day – 6 Introduction to network theory

Task – 1 Listing the functions of ports.

Device	Main Functions	Typical Port Types & Count
Router	Routes traffic between your home network and the Internet (WAN); assigns IP addresses (DHCP); performs Network Address Translation (NAT).	1 WAN Port (Ethernet/Coax/Fiber) + 4 LAN Ports (RJ45 Ethernet).
Switch	Connects multiple wired devices on the same local network (LAN) and routes traffic directly using MAC addresses.	4 to 8+ LAN Ports (RJ45 Ethernet).
Hub	Broadcasts incoming data to all connected devices indiscriminately (legacy device, rarely used today).	4 to 8 LAN Ports (RJ45 Ethernet).
Firewall	Inspects incoming/outgoing network traffic and blocks unauthorized access based on security rules (usually built into the home router).	Virtual/Software (filters traffic passing between WAN and LAN ports).
Access Point (AP)	Transmits wireless signals (Wi-Fi) to connect wireless devices to the wired network (often built into home routers).	1 LAN Port (RJ45 Ethernet, often with PoE) for upstream uplink.

Task – 2 Diagram, Labelling and devices.

- 1. Wireless Wi-Fi Router
- WAN / Internet Port (Ethernet): Usually colored yellow or blue and labeled "WAN" or "Internet". This connects to your modem.
- LAN Ports (Ethernet): Usually grouped together (labeled 1 to 4) in yellow or black. These connect wired devices like PCs and consoles.
- Antennas: External adjustable rods (or internal modules) for 2.4 GHz / 5 GHz Wi-Fi signals.
- Power Port: A small round DC barrel jack labeled "Power" or "12V DC" next to a power button.
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- 2. Enterprise Router / Switch (e.g., Cisco)
- Console Port: Often colored light blue (RJ45 or Mini-USB format). Used for initial command-line setup via a computer.
- Ethernet Ports (FastEthernet / GigabitEthernet): Standard RJ45 sockets used for local network traffic and inter-switch links.
- Power Port: A 3-prong AC power connector (C14 socket) for a standard computer power cord.
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- 3. Standalone Wireless Access Point (AP)
- Ethernet / PoE Port: An RJ45 port labeled "ETH" or "PoE" (Power over Ethernet) that delivers both network data and electrical power through a single cable.
- Console Port: A small RJ45 or micro-USB port used by administrators for local management.
- External Antennas: Screw-on RP-SMA connectors where high-gain Wi-Fi antennas attach.

Task – 3 Major role on the device in the app.

- When you open Instagram on my phone:
- Wireless Access Point (AP): Converts your phone's Wi-Fi signals into physical electrical signals and sends them over an Ethernet cable to your local network.
- Router: Directs your request (e.g., loading your feed or reels) out of your private home network onto the public Internet, routing it through your ISP to Instagram's server IP address

- Firewall: Inspects incoming data packets returning from Instagram's servers to ensure they are safe and authorized, blocking any unauthorized or suspicious traffic from reaching your device.

Task – 4 Reason of slow wifi and disconnections.

- The issue is likely caused by overloaded Wireless Access Points (APs). In a college environment, a single AP often tries to handle hundreds of simultaneous client connections (phones, laptops, tablets). When an AP exceeds its client capacity or bandwidth limits, it creates high signal interference, extreme latency, and drops active connections to clear its buffer.
- Upgrade Target: Upgrade to high-density, Wi-Fi 6 / 6E Access Points with Multi-User MIMO (MU-MIMO) support.
- Modern high-density APs can handle 200+ active devices simultaneously without dropping packets.
- MU-MIMO allows the AP to communicate with multiple devices at the exact same time instead of forcing devices to wait in line, eliminating slow speeds and unexpected disconnections.
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